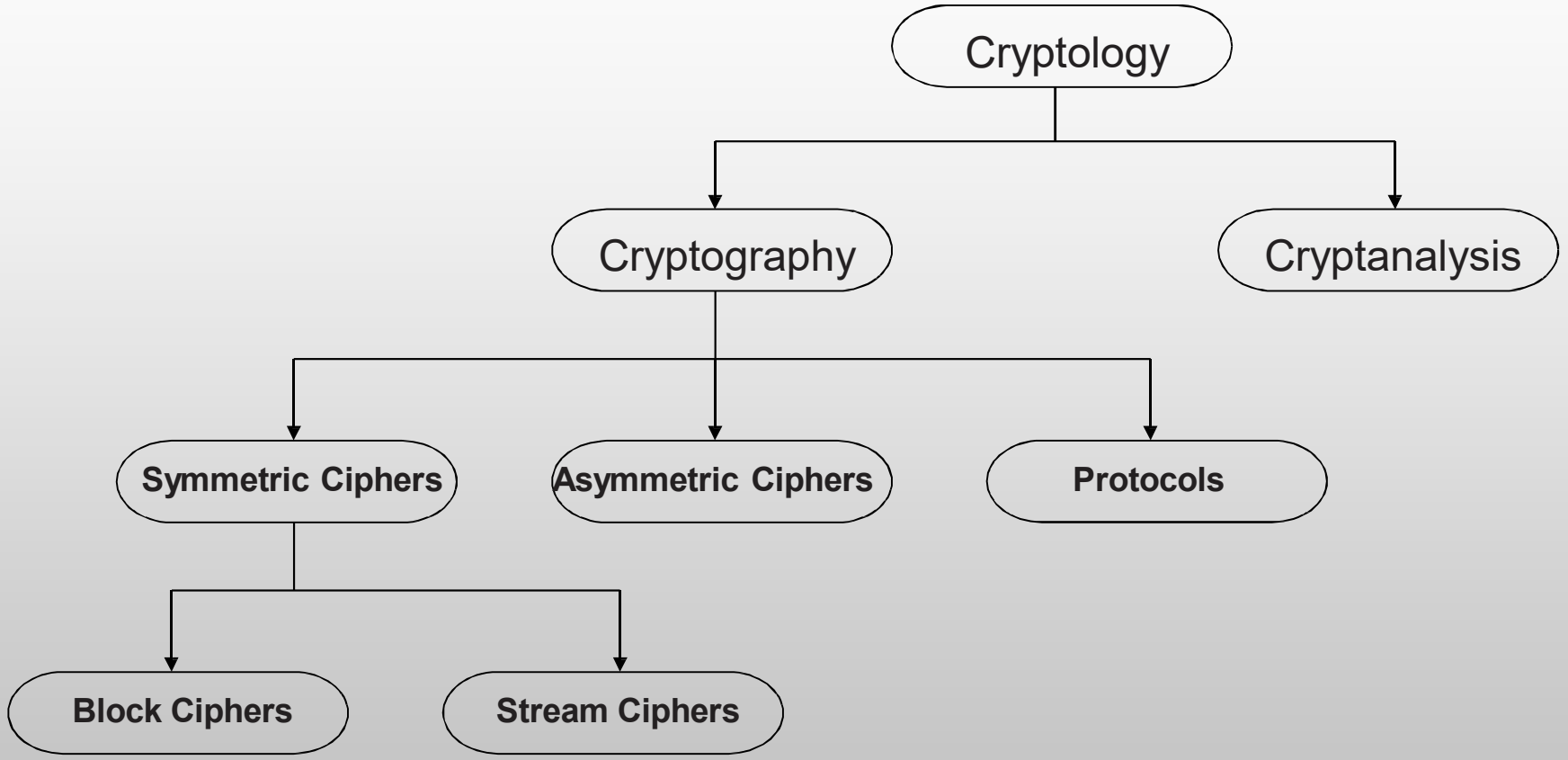
INTRODUCTION TO CRYPTOGRAPHY



Classification of the Field of Cryptology



Some Basic Facts

- **Ancient Crypto:** Early signs of encryption in Eqypt in ca. 2000 B.C.
  Letter-based encryption schemes (e.g., Caesar cipher) popular ever since.

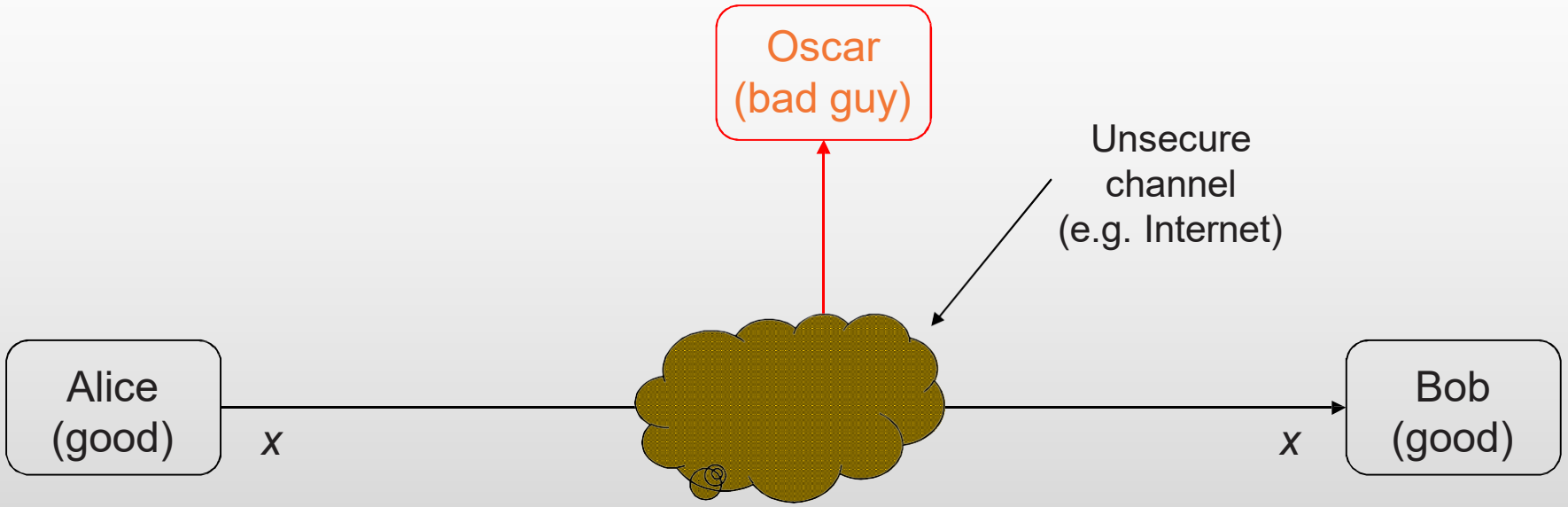
- **Symmetric ciphers:** All encryption schemes from ancient times until 1976 were symmetric ones.

- **Asymmetric ciphers:** In 1976 public-key (or asymmetric) cryptography was openly proposed by Diffie, Hellman and Merkle.

- **Hybrid Schemes:** The majority of today‘s protocols are hybrid schemes, i.e., the use both
  - symmteric ciphers (e.g., for encryption and message authentication) and
  - asymmetric ciphers (e.g., for key exchange and digital signature).

Symmetric Cryptography

- Alternative names: **private-key**, **single-key** or **secret-key** cryptography.

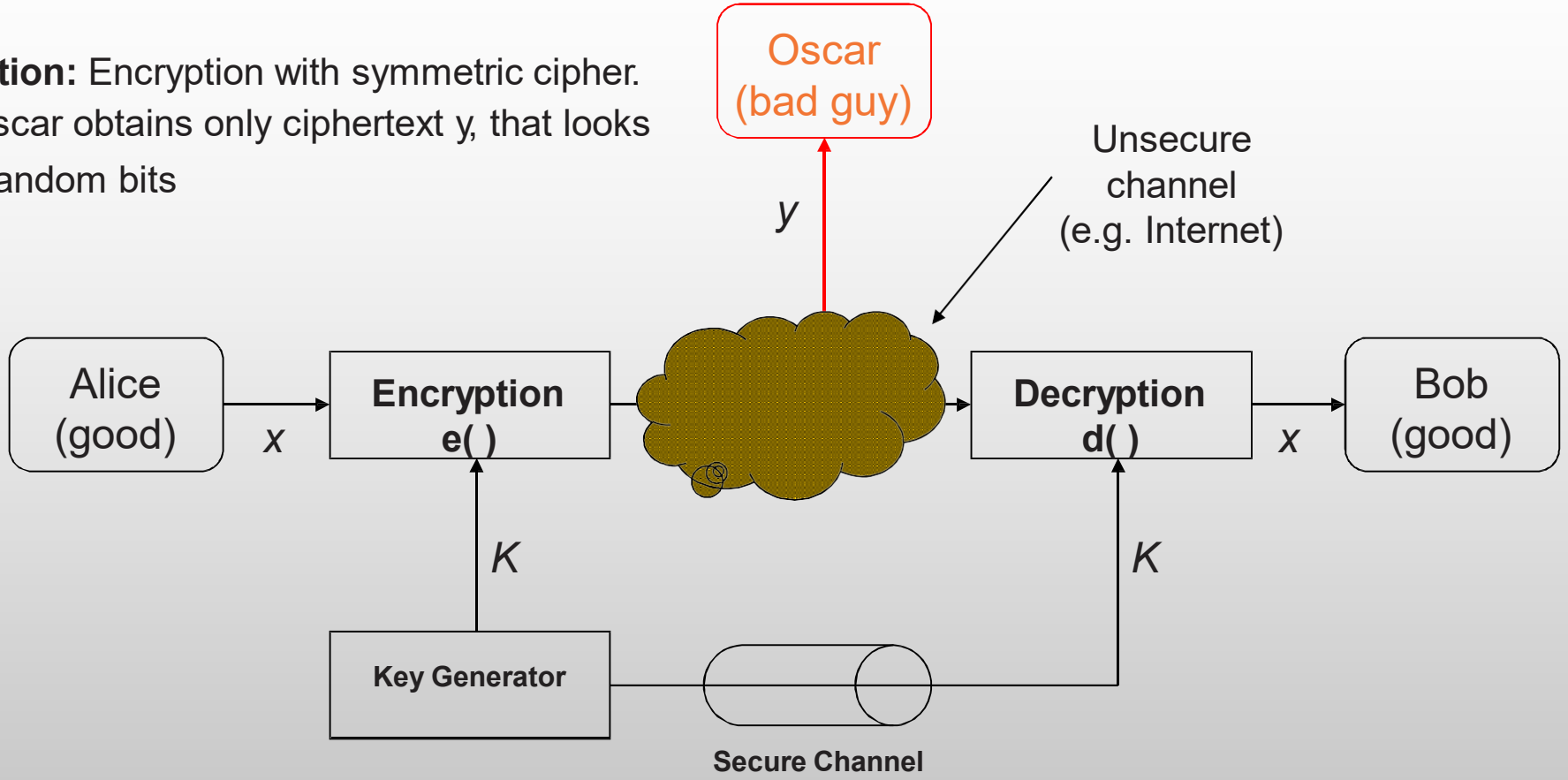


- **Problem Statement:**
  1)Alice and Bob would like to communicate via an unsecure channel (e.g., WLAN or Internet).

  2)A malicious third party Oscar (the bad guy) has channel access but should not be able to understand the communication.

Symmetric Cryptography

Solution: Encryption with symmetric cipher.
⇒ Oscar obtains only ciphertext y, that looks like random bits



- x is the **plaintext**
- y is the **cipher-text**
- *K* is the **key**
- Set of all keys {*K*1, *K*2, ...,*Kn*} is the **key space**

Symmetric Cryptography

- Encryption equation $\quad y = e_K(x)$
- Decryption equation $\quad x = d_K(y)$

- Encryption and decryption are inverse operations if the same key K is used on both sides:

$$d_K(y) = d_K(e_K(x)) = x$$

- Important: The key must be transmitted via a **secure channel** between Alice and Bob.
- The secure channel can be realized, e.g., by manually installing the key for the Wi-Fi Protected Access (WPA) protocol or a human courier.
- However, the system is only secure if an attacker does not learn the key K!

$\Rightarrow$ **The problem of secure communication is reduced to secure transmission and storage of the key K.**

Cryptanalysis

- There is no *mathematical proof of security* for any practical cipher
- The only way to have assurance that a cipher is secure is to try to break it (and fail) !

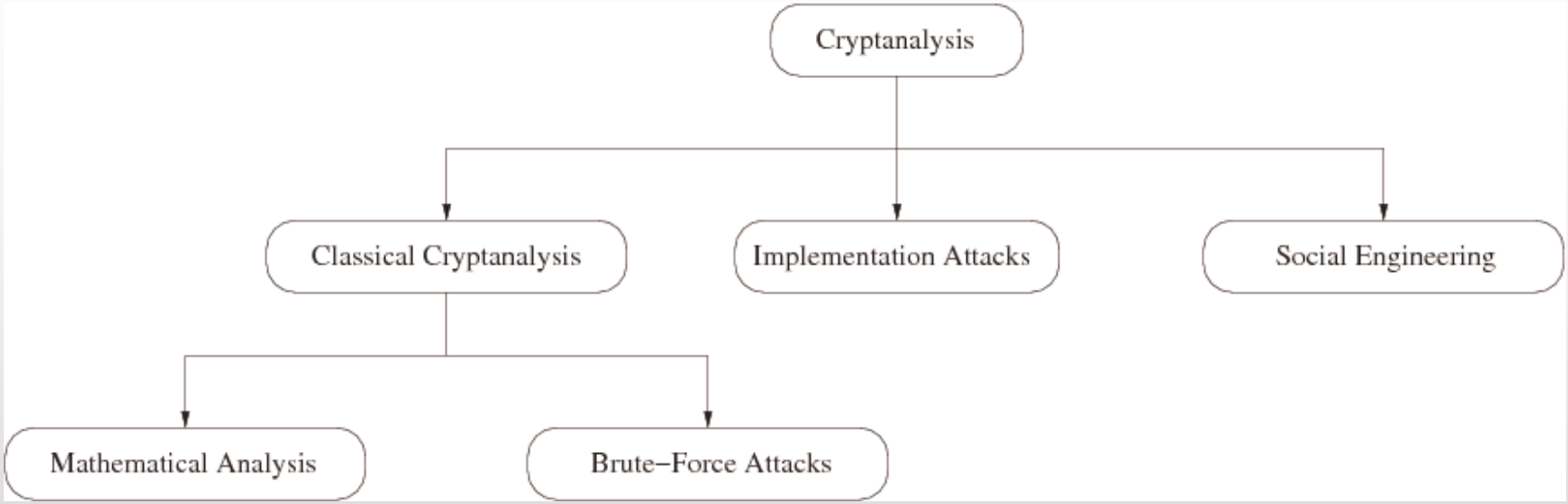
Kerckhoff's Principle is paramount in modern cryptography:

> A cryptosystem should be secure even if the attacker (Oscar) knows **all details** about the system, with the exception of the **secret key.**

- In order to achieve Kerckhoff's Principle in practice:
  Only use widely known ciphers that have been cryptanalyzed for several years by good cryptographers!

  Remark: It is tempting to assume that a cipher is „more secure" if its details are kept secret. However, history has shown time and again that secret ciphers can almost always been broken once they have been reversed engineered. (Example: Content Scrambling System (CSS) for DVD content protection.)

Cryptanalysis: Attacking Cryptosystems



- **Classical Attacks**
  - Mathematical Analysis
  - Brute-Force Attack
- **Implementation Attack**: Try to extract key through reverese engineering or power measurement, e.g., for a banking smart card.
- **Social Engineering**: E.g., trick a user into giving up her password

Brute-Force Attack (or Exhaustive Key Search) against Symmetric Ciphers

- Treats the cipher as a black box
- Requires (at least) 1 plaintext-ciphertext pair (x_0, y_0)
- Check all possible keys until condition is fulfilled:

$$d_K(y_0) \stackrel{?}{=} x_0$$

- How many keys do we need ?

Key length in bit	Key space	Security life time (assuming brute-force as best possible attack)
64	2^{64}	**Short term** (few days or less)
128	2^{128}	**Long-term** (several decades in the absence of quantum computers)
256	2^{256}	**Long-term** (also resistant against quantum computers – note that QC do not exist at the moment and might never exist)



Substitution Cipher

- Historical cipher
- Great tool for understanding brute-force vs. analytical attacks
- Encrypts letters rather than bits (like all ciphers until after WW II)

Idea: replace each plaintext letter by a fixed other letter.

Plaintext		Ciphertext
A	→	k
B	→	d
C	→	w
	....	

for instance, `ABBA` would be encrypted as `kddk`

- Example (ciphertext):

```
iq ifcc vqqr fb rdq vfllcq na rdq cfjwhwz hr bnnb  hcc
                hwwhbsqvqbre hwq vhlq
```

- How secure is the Substitution Cipher? Let‘s look at attacks…

Attacks against the Substitution Cipher

1. **Attack: Exhaustive Key Search (Brute-Force Attack)**

- Simply try every possible subsititution table until an intelligent plaintext appears (note that each substitution table is a key)..
- How many substitution tables (= keys) are there?

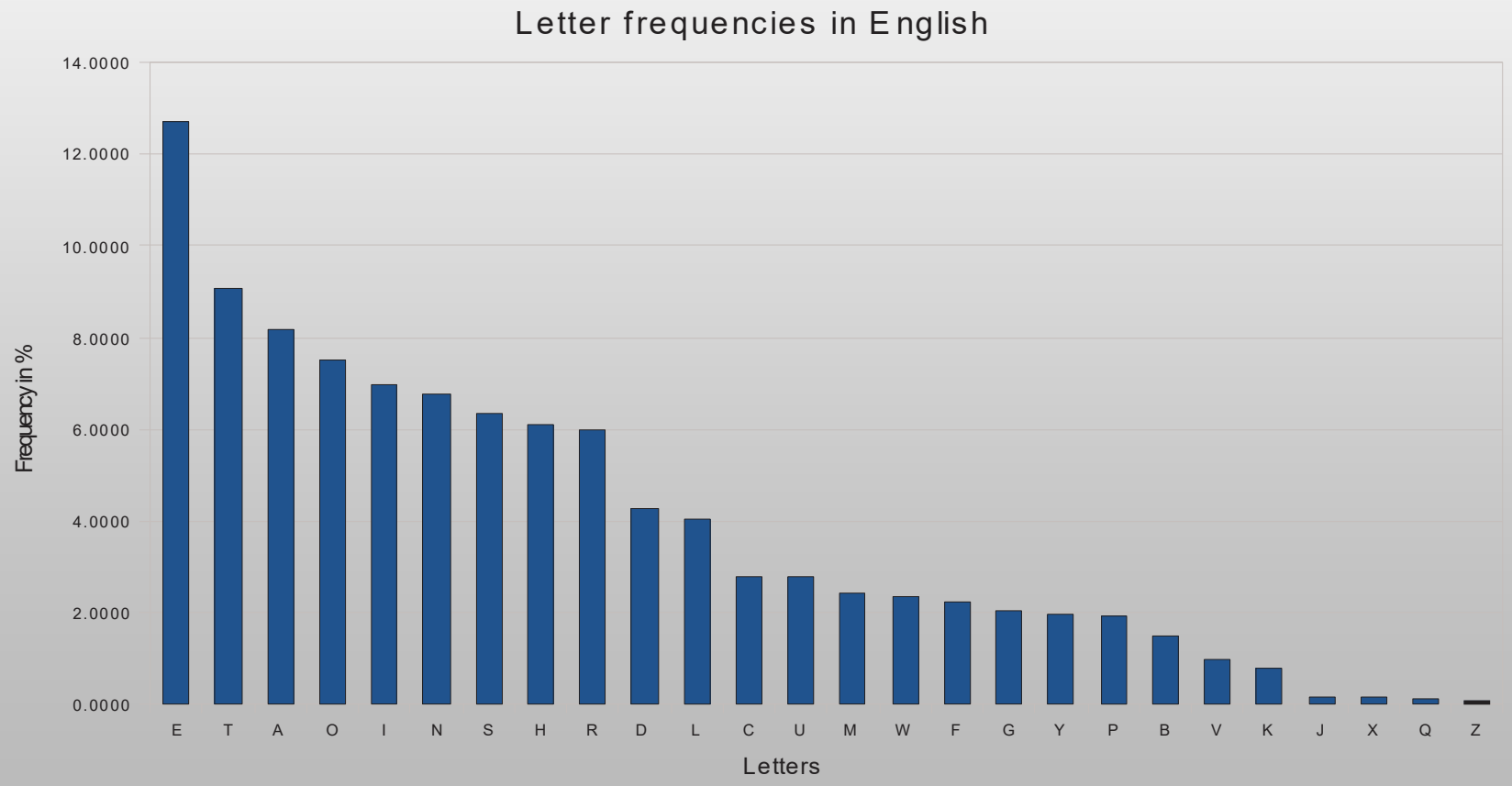
$$26 \times 25 \times \ldots \times 3 \times 2 \times 1 = 26! \approx 2^{88}$$

Search through 2^{88} keys is completely infeasible with today's computers!
(cf. earlier table on key lengths)

- Q: Can we now conclude that the substitution cipher is secure since a brute-forece attack is not feasible?
- A: No! We have to protect against **all** possible attacks…

2. Attack: Letter Frequency Analysis (Brute-Force Attack)

- Letters have very different frequencies in the English language
- Moreover: the frequency of plaintext letters is preserved in the ciphertext.
- For instanc, „e“ is the most common letter in English; almost 13% of all letters in a typical English text are „e“.
- The next most common one is „t“ with about 9%.



Breaking the Substitution Cipher with Letter Frequency Attack

- Let‘s retun to our example and identify the most frequent letter:

```
iq ifcc vqqr fb rdq vfllcq na rdq cfjwhwz hr bnnb hcc
hwwhbsqvqbre hwq vhlq
```

- We replace the ciphertext letter `q` by `E` and obtain:

```
iE ifcc vEEr fb rdE vfllcE na rdE cfjwhwz hr bnnb  hcc
hwwhbsEvEbre hwE vhlE
```

- By further guessing based on the frequency of the remaining letters we obtain the plaintext:

```
WE WILL MEET IN THE MIDDLE OF THE LIBRARY AT NOON ALL
ARRANGEMENTS ARE MADE
```

- In practice, not only frequencies of individual letters can be used for an attack, but also the frequency of letter pairs (i.e., „th“ is very common in English), letter triples, etc.

CAESAR CIPHER

- WHEN JULIUS CAESAR SENT MESSAGES TO HIS GENERALS, HE DIDN'T TRUST HIS MESSENGERS.

- HE ENCRYPTED HIS MESSAGES BY REPLACING EVERY LETTER:
  - A WITH A D
  - B WITH AN E
  - AND SO ON

- HIS GENERALS KNEW THE "SHIFT BY 3" RULE AND COULD DECIPHER HIS MESSAGES.

- SHIFT BY ‘N’ = SHIFT CIPHER.



Vigenere Table (Tabula recta)

Plain Text

Key

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y



Blaise de Vigenère

Plaintext: ATTACKATDAWN
Key: LEMONLEMONLE
Ciphertext: LXFOPVEFRNHR

Period: ?

Columnar Transposition

- This method rearranges plaintext by rows/columns
- Example: THIS IS A SECRET MESSAGE
- Arranged in rows (ignoring word separations):

  THIS
  ISAS
  ECRE
  TMES
  SAGE
- Reading by columns, the resulting ciphertext is

  TIETSHSCMAIAREGSSESE
- Omitting spaces makes ciphertext even less readable

ENIGMA MACHINE



- Is an electric-mechanical machine generating ciphers
- Commercially available first in 1920
- Adopted by a number of nations for government/military purposes
- Most famous use by Nazi Germany
- Was cracked by Allied cryptologists
- Fascinating history (Bletchley Park) – ‘The Imitation Game’
- Designed in 1918, adapted and improved by German army by 1928
- Allies managed to get hold of one machine
  - This helped to break code (and probably helped finishing off the war earlier)

The ENIGMA Machine

- Combination of mechanical and electrical systems
- Keyboard is used to type plain text message
- Rotating discs (“rotors”) form varying electrical circuits
- Electrical current flows to output lamp
- This indicates resulting ciphertext
- High security because of constant change of rotor positions

- ENIGMA Explained